

PeriDEM – High-fidelity modeling of granular media consisting of deformable complex-shaped particles

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

Accurate simulation of granular materials under extreme mechanical conditions, such as crushing, fracture, and large deformation, remains a significant challenge in geotechnical, manufacturing, and mining applications. Classical discrete element method (DEM) models typically treat particles as rigid or nearly rigid bodies, limiting their ability to capture internal deformation and fracture. The PeriDEM library, first introduced by Jha et al. (2021), addresses this limitation by modeling particles as deformable solids using peridynamics, a nonlocal continuum theory that naturally accommodates fracture and significant deformation. Inter-particle contact is handled using DEM-inspired local laws, enabling realistic interaction between complex-shaped particles.

Implemented in C++, PeriDEM is designed for extensibility and ease of deployment. It relies on a minimal set of external libraries, supports multithreaded execution, and includes demonstration examples involving compaction, fracture, and rotational dynamics. The framework facilitates granular-scale simulations, supports the development of constitutive models, and serves as a foundation for multi-fidelity coupling in real-world applications.

Statement of Need

Granular materials play a central role in many engineered systems, but modeling their behavior under high loading, deformation, and fragmentation remains an open problem. Popular open-source DEM codes such as YADE (Smilauer et al., 2021), BlazeDEM (Govender et al., 2016), Chrono DEM-Engine (Zhang et al., 2024), and LAMMPS (Thompson et al., 2022) are widely used but typically treat particles as rigid, limiting their accuracy in scenarios involving internal deformation and breakage. A recent review by Dosta et al. (2024) compares several DEM libraries. Meanwhile, peridynamics-based codes such as Peridigm (Littlewood et al., 2024) and NLMech (Jha & Diehl, 2021) are designed to simulate deformation and fracture within a single structure, with limited support for multi-structure simulations.

PeriDEM fills this gap by integrating state-based peridynamics for intra-particle deformation with DEM-style contact laws for particle interactions. This hybrid approach enables direct simulation of particle fragmentation, stress redistribution, and dynamic failure propagation—capabilities essential for modeling granular compaction, attrition, and crushing.

Recent multiscale approaches, including DEM-continuum and DEM-level-set coupling methods (Harmon et al., 2021), aim to bridge scales but often rely on homogenization assumptions. Sand crushing in geotechnical systems, for example, has been modeled using micro-CT-informed FEM or phenomenological laws (Chen et al., 2023). PeriDEM offers a particle-resolved alternative that allows bottom-up investigation of granular failure and shape evolution, especially in systems where fragment dynamics are critical.

41 Brief Introduction to PeriDEM Model

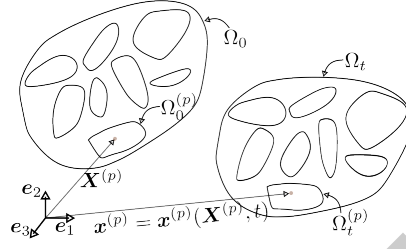


Figure 1: Motion of particle system.

42 Consider a fixed frame of reference and $\{e_i\}_{i=1}^d$ are orthonormal bases. Consider a collection
 43 of N_P particles $\Omega_0^{(p)}$, $1 \leq p \leq N_P$, where $\Omega_0^{(p)} \subset \mathbb{R}^d$ with $d = 2, 3$ represents the initial
 44 configuration of particle p . Suppose $\Omega_0 \supset \bigcup_{p=1}^{N_P} \Omega_0^{(p)}$ is the domain containing all particles; see
 45 Figure 1. The particles in Ω_0 are dynamically evolving due to external boundary conditions and
 46 internal interactions; let $\Omega_t^{(p)}$ denote the configuration of particle p at time $t \in (0, t_F]$, and
 47 $\Omega_t \supset \bigcup_{p=1}^{N_P} \Omega_t^{(p)}$ domain containing all particles at that time. The motion $x^{(p)} = x^{(p)}(X^{(p)}, t)$
 48 takes point $X^{(p)} \in \Omega_0^{(p)}$ to $x^{(p)} \in \Omega_t^{(p)}$, and collectively, the motion is given by $x = x(X, t) \in$
 49 Ω_t for $X \in \Omega_0$. We assume the media is dry and not influenced by factors other than
 50 mechanical loading (e.g., moisture and temperature are not considered). The configuration of
 51 particles in Ω_t at time t depends on various factors, such as material and geometrical properties,
 52 contact mechanism, and external loading. Essentially, there are two types of interactions
 53 present in the media:

- 54 ■ *Intra-particle interaction* that models the deformation and internal forces in the particle
 55 and
- 56 ■ *Inter-particle interaction* that accounts for the contact between particles and the boundary
 57 of the domain in which the particles are contained.

58 In DEM, the first interaction is ignored, assuming that particle deformation is insignificant
 59 compared to inter-particle interactions. On the other hand, PeriDEM accounts for both
 60 interactions.

61 The balance of linear momentum for particle p , $1 \leq p \leq N_P$, takes the form:

$$\rho^{(p)} \ddot{\mathbf{u}}^{(p)}(X, t) = \mathbf{f}_{int}^{(p)}(X, t) + \mathbf{f}_{ext}^{(p)}(X, t), \quad \forall (X, t) \in \Omega_0^{(p)} \times (0, t_F), \quad (1)$$

62 where $\rho^{(p)}$, $\mathbf{f}_{int}^{(p)}$, and $\mathbf{f}_{ext}^{(p)}$ are density, and internal and external force densities. The
 63 above equation is complemented with initial conditions, $\mathbf{u}^{(p)}(X, 0) = \mathbf{u}_0^{(p)}(X)$, $\dot{\mathbf{u}}^{(p)}(X, 0) =$
 64 $\dot{\mathbf{u}}_0^{(p)}(X)$, $X \in \Omega_0^{(p)}$.

65 Internal force – State-based peridynamics

66 The internal force term $\mathbf{f}_{int}^{(p)}(X, t)$ in the momentum balance governs intra-particle deformation
 67 and fracture. In PeriDEM, this term is modeled using a simplified state-based peridynamics
 68 formulation that accounts for nonlocal interactions over a finite horizon. The underlying model
 69 and its numerical implementation are discussed in detail by Jha et al. (2021).

70 DEM-inspired contact forces

71 The external force term $\mathbf{f}_{ext}^{(p)}(X, t)$ includes body forces, wall-particle interactions, and contact
 72 forces from other particles. Contact is modeled using a spring-dashpot-slider formulation

73 applied locally when particles come within a critical distance; see Figure 2. This approach
74 introduces nonlinear normal forces, damping, and friction without relying on particle convexity
75 or geometric simplifications. The full formulation of contact detection, force assembly, and
76 implementation is detailed by Jha et al. (2021).

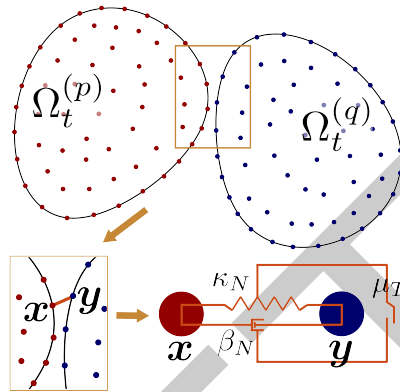


Figure 2: High-resolution contact approach in the PeriDEM model for granular materials between arbitrarily shaped particles. The spring dashpot slider system represents normal contact elasticity, normal damping, and tangential friction between material points x and y . Adapted from Jha et al. (2021).

77 Implementation

78 PeriDEM is implemented in C++ and hosted on GitHub. It depends on a minimal set of external
79 libraries, most of which are bundled in the external directory. Key dependencies include
80 Taskflow (Huang et al., 2021) for multithreaded parallelism, nanoflann (Blanco & Rai, 2014)
81 for efficient neighborhood search, and VTK for output. The numerical strategies for neighbor
82 search, peridynamic integration, damage evaluation, and time stepping follow those introduced
83 by Jha et al. (2021). The core simulation model is implemented in `src/model/dem`, with the
84 class `DEMModel` managing particle states, force calculations, and time integration. This work
85 builds on earlier research in the analysis and numerical methods for peridynamics (Jha et al.,
86 2025; Jha & Lipton, 2018a, 2018b, 2019; Jha & Lipton, 2020).

87 Features

- 88 ■ Combines peridynamics and DEM to model intra-particle deformation and inter-particle
89 contact
- 90 ■ Simulates deformation and breakage of single particles under complex loading conditions
- 91 ■ Supports arbitrarily shaped particles for realistic granular systems
- 92 ■ Ongoing development of MPI-based parallelism and adaptive modeling strategies to
93 improve efficiency without sacrificing accuracy
- 94 ■
- 95 ■
- 96 ■

97 Examples

98 Example cases are described in [examples/README.md](#). One key simulation demonstrates
99 the compression of over 500 circular and hexagonal particles in a rectangular container by
100 displacing the top wall. The stress on the wall becomes increasingly nonlinear with penetration
101 depth as damage accumulates and the medium yields (see Figure 3a).

102 Preliminary performance tests show that compute time increases exponentially with particle
103 count due to the nonlocal nature of both peridynamic and contact interactions—highlighting

a computational bottleneck. This motivates future integration of MPI-based parallelism and a multi-fidelity modeling framework. Additional examples include attrition of non-circular particles in a rotating cylinder (Figure 3c).

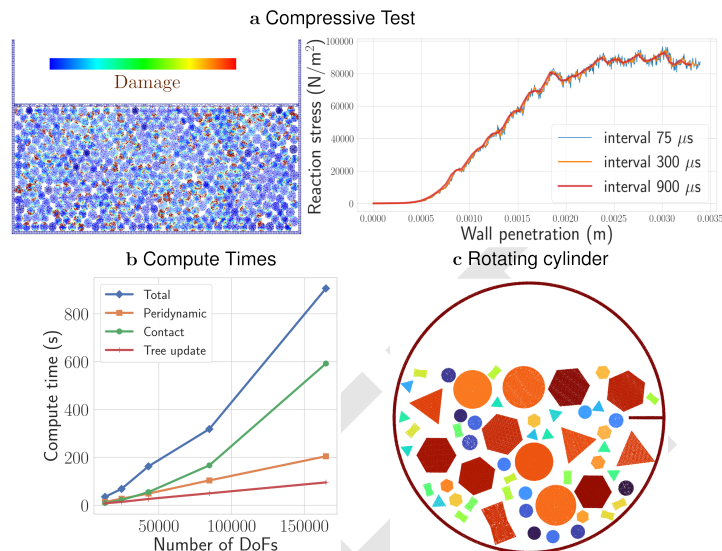


Figure 3: (a) Nonlinear response under compression, (b) exponential growth of compute time due to nonlocality of internal and contact forces, and (c) rotating cylinder with nonspherical particles. Adapted from Jha et al. (2021).

Acknowledgements

Current efforts on the development and enhancement of the PeriDEM library are supported by the U.S. National Science Foundation through the Engineering Research Initiation (ERI) program under Grant No. 2502279.

References

- Blanco, J. L., & Raj, P. K. (2014). *Nanoflann: A C++ header-only fork of FLANN, a library for nearest neighbor (NN) with KD-trees*. <https://github.com/jlblancoc/nanoflann>.
- Chen, Q., Li, Z., Dai, Z., Wang, X., Zhang, C., & Zhang, X. (2023). Mechanical behavior and particle crushing of irregular granular material under high pressure using discrete element method. *Scientific Reports*, 13(1), 7843. <https://doi.org/10.1038/s41598-023-35022-w>
- Dosta, M., André, D., Angelidakis, V., Caulk, R. A., Celigueta, M. A., Chareyre, B., Dietiker, J.-F., Girardot, J., Govender, N., Hubert, C., & others. (2024). Comparing open-source DEM frameworks for simulations of common bulk processes. *Computer Physics Communications*, 296, 109066. <https://doi.org/10.1016/j.cpc.2023.109066>
- Govender, N., Wilke, D. N., & Kok, S. (2016). Blaze-DEMGPU: Modular high performance DEM framework for the GPU architecture. *SoftwareX*, 5, 62–66. <https://doi.org/10.1016/j.softx.2016.04.004>
- Harmon, J. M., Karapiperis, K., Li, L., Moreland, S., & Andrade, J. E. (2021). Modeling connected granular media: Particle bonding within the level set discrete element method. *Computer Methods in Applied Mechanics and Engineering*, 373, 113486. <https://doi.org/10.1016/j.cma.2020.113486>

- 128 Huang, T.-W., Lin, D.-L., Lin, C.-X., & Lin, Y. (2021). Taskflow: A lightweight parallel and
129 heterogeneous task graph computing system. *IEEE Transactions on Parallel and Distributed*
130 *Systems*, 33(6), 1303–1320. <https://doi.org/10.1109/TPDS.2021.3104255>
- 131 Jha, P. K., Desai, P. S., Bhattacharya, D., & Lipton, R. (2021). Peridynamics-based discrete
132 element method (PeriDEM) model of granular systems involving breakage of arbitrarily
133 shaped particles. *Journal of the Mechanics and Physics of Solids*, 151, 104376. <https://doi.org/10.1016/j.jmps.2021.104376>
- 134
- 135 Jha, P. K., & Diehl, P. (2021). NLMech: Implementation of finite difference/meshfree
136 discretization of nonlocal fracture models. *Journal of Open Source Software*, 6(65), 3020.
137 <https://doi.org/10.21105/joss.03020>
- 138 Jha, P. K., Diehl, P., & Lipton, R. (2025). Nodal finite element approximation of peridynamics.
139 *Computer Methods in Applied Mechanics and Engineering*, 434, 117519. <https://doi.org/10.1016/j.cma.2024.117519>
- 140
- 141 Jha, P. K., & Lipton, R. (2018a). Numerical analysis of nonlocal fracture models in holder
142 space. *SIAM Journal on Numerical Analysis*, 56(2), 906–941. <https://doi.org/10.1137/17M1112236>
- 143
- 144 Jha, P. K., & Lipton, R. (2018b). Numerical convergence of nonlinear nonlocal continuum
145 models to local elastodynamics. *International Journal for Numerical Methods in Engineering*,
146 114(13), 1389–1410. <https://doi.org/10.1002/nme.5791>
- 147 Jha, P. K., & Lipton, R. (2019). Numerical convergence of finite difference approximations for
148 state based peridynamic fracture models. *Computer Methods in Applied Mechanics and*
149 *Engineering*, 351, 184–225. <https://doi.org/10.1016/j.cma.2019.03.024>
- 150 Jha, P. K., & Lipton, R. P. (2020). Kinetic relations and local energy balance for LEFM from
151 a nonlocal peridynamic model. *International Journal of Fracture*. <https://doi.org/10.1007/s10704-020-00480-0>
- 152
- 153 Littlewood, D. J., Parks, M. L., Foster, J. T., Mitchell, J. A., & Diehl, P. (2024). The
154 peridigm meshfree peridynamics code. *Journal of Peridynamics and Nonlocal Modeling*,
155 6(1), 118–148. <https://doi.org/10.1007/s42102-023-00100-0>
- 156 Smilauer, V., Angelidakis, V., Catalano, E., Caulk, R., Chareyre, B., Chèvremont, W., Doro-
157 feenko, S., Duriez, J., Dyck, N., Elias, J., Er, B., Eulitz, A., Gladky, A., Guo, N., Jakob, C.,
158 Kneib, F., Kozicki, J., Marzougui, D., Maurin, R., ... Yuan, C. (2021). Yade documentation
159 (3rd edition). *Zenodo*. <https://doi.org/10.5281/zenodo.5705394>
- 160 Thompson, A. P., Aktulga, H. M., Berger, R., Bolintineanu, D. S., Brown, W. M., Crozier, P.
161 S., in 't Veld, P. J., Kohlmeyer, A., Moore, S. G., Nguyen, T. D., Shan, R., Stevens, M. J.,
162 Tranchida, J., Trott, C., & Plimpton, S. J. (2022). LAMMPS - a flexible simulation tool for
163 particle-based materials modeling at the atomic, meso, and continuum scales. *Computer*
164 *Physics Communications*, 271, 108171. <https://doi.org/10.1016/j.cpc.2021.108171>
- 165 Zhang, R., Tagliafierro, B., Vanden Heuvel, C., Sabarwal, S., Bakke, L., Yue, Y., Wei, X.,
166 Serban, R., & Negruț, D. (2024). Chrono DEM-Engine: A discrete element method dual-
167 GPU simulator with customizable contact forces and element shape. *Computer Physics*
168 *Communications*, 300, 109196. <https://doi.org/10.1016/j.cpc.2024.109196>